MTH636 – Game Theory

Lecture 15: Nash's Theorem Proof & Extensive Form Games

1 Proof of Nash's Theorem (Continued)

Theorem: Nash's Existence Theorem

For every finite game, there is a Nash Equilibrium in mixed strategies.

Proof (Contd.): Let $G = \langle N, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$. Let $|N| = n$ and $|S_i| = m_i$ for all $i = 1, \ldots, n$.

Recall the gain function $g_i^j : \Sigma \to \mathbb{R}$ defined for each player i and pure strategy s_i^j:

$$g_i^j(\sigma) = \max\left\{0, U_i(s_i^j, \sigma_{-i}) - U_i(\sigma)\right\}$$

Claim 1: g_i^j is a continuous function of σ for all i and j.
Claim 2: σ is an NE if and only if $g_i^j(\sigma) = 0$ for all $i = 1, \ldots, n$ and $j = 1, \ldots, m_i$.

Our Goal: We want to construct a mapping $f : \Sigma \to \Sigma$ such that:

- f is continuous.
- Every fixed point of f is an NE of the game.

Then by Brouwer's Fixed Point Theorem, f will have a fixed point, proving an NE exists.

Construction of f: Define $f = (f_1, \ldots, f_n)$ where $f_i = (f_i^1, \ldots, f_i^{m_i})$ is given by:

$$f_i^j(\sigma) = \frac{\sigma_i(s_i^j) + g_i^j(\sigma)}{1 + \sum_{k=1}^{m_i} g_i^k(\sigma)} \quad \forall i = 1, \ldots, n, \ \forall j = 1, \ldots, m_i$$

Claim 3: $f(\sigma) \in \Sigma$.

Proof. We must verify two properties for $f_i(\sigma)$ to be a valid probability distribution:

1. $f_i^j(\sigma) \geq 0$: This holds because $\sigma_i(s_i^j) \geq 0$ and $g_i^j(\sigma) \geq 0$.
2. $\sum_{j=1}^{m_i} f_i^j(\sigma) = 1$:

$$\sum_{j=1}^{m_i} \frac{\sigma_i(s_i^j) + g_i^j(\sigma)}{1 + \sum_{k=1}^{m_i} g_i^k(\sigma)} = \frac{\sum_{j=1}^{m_i} \sigma_i(s_i^j) + \sum_{j=1}^{m_i} g_i^j(\sigma)}{1 + \sum_{k=1}^{m_i} g_i^k(\sigma)} = \frac{1 + \sum_{j=1}^{m_i} g_i^j(\sigma)}{1 + \sum_{k=1}^{m_i} g_i^k(\sigma)} = 1$$

□

Claim 4: f is a continuous function of σ.
Proof: This follows directly since the components $g_i^j(\sigma)$ are continuous functions.

Claim 5: Let σ be a fixed point of f. Then $g_i^j(\sigma) = \sigma_i(s_i^j) \sum_{k=1}^{m_i} g_i^k(\sigma)$.

Proof. Suppose σ is a fixed point, meaning $f_i^j(\sigma) = \sigma_i(s_i^j)$. Then:

$$\frac{\sigma_i(s_i^j) + g_i^j(\sigma)}{1 + \sum_{k=1}^{m_i} g_i^k(\sigma)} = \sigma_i(s_i^j)$$

$$\sigma_i(s_i^j) + g_i^j(\sigma) = \sigma_i(s_i^j) + \sigma_i(s_i^j) \sum_{k=1}^{m_i} g_i^k(\sigma)$$

$$\implies g_i^j(\sigma) = \sigma_i(s_i^j) \sum_{k=1}^{m_i} g_i^k(\sigma)$$

□

Claim 6: If σ is a fixed point, then σ is an NE of G.

Proof. Assume for contradiction that σ is a fixed point but **not** an NE. This implies there exists some player $p \in N$ and strategy $s_p^k \in S_p$ such that $g_p^k(\sigma) > 0$. Consequently, the sum of gains for player p is strictly positive: $\sum_{k=1}^{m_p} g_p^k(\sigma) > 0$.

As σ is a fixed point, by Claim 5:

$$g_p^j(\sigma) = \sigma_p(s_p^j) \sum_{k=1}^{m_p} g_p^k(\sigma)$$

Since the sum is strictly positive, we have:

$$g_p^j(\sigma) > 0 \iff \sigma_p(s_p^j) > 0 \quad \forall j = 1, \ldots, m_p$$

Now consider Player p's expected utility:

$$U_p(\sigma) = \sum_{k=1}^{m_p} \sigma_p(s_p^k) U(s_p^k, \sigma_{-p})$$

Subtract $U_p(\sigma)$ from both sides (using the fact that $\sum \sigma_p(s_p^k) = 1$):

$$0 = \sum_{k=1}^{m_p} \sigma_p(s_p^k) \left(U(s_p^k, \sigma_{-p}) - U_p(\sigma) \right)$$

Since $g_p^k(\sigma) > 0$ for any s_p^k in the support, we know $U(s_p^k, \sigma_{-p}) - U_p(\sigma)$ is exactly equal to $g_p^k(\sigma)$. So:

$$\sum_{s_p^k | \sigma_p(s_p^k) > 0} \sigma_p(s_p^k) g_p^k(\sigma) = 0$$

However, since $\sigma_p(s_p^k) > 0 \implies g_p^k(\sigma) > 0$, we are summing strictly positive terms, which cannot sum to 0. This is a contradiction.

Thus, σ must be a Nash Equilibrium. By Brouwer's theorem, such a fixed point exists, so a mixed NE exists. □

2 Extensive Form Games: Introduction

We now model games where players take turns sequentially.

Example: Sequential Battle of the Sexes

The man takes his action first, and then the woman sees his choice and makes her own move.

Game Tree:

- Root node (M): Man chooses F (Football) or D (Dance).
- If M chooses F, we reach node W_1. Woman chooses F or D. Payoffs: $(2,1)$ for FF, $(0,0)$ for FD.
- If M chooses D, we reach node W_2. Woman chooses F or D. Payoffs: $(0,0)$ for DF, $(1,2)$ for DD.

Strategies: A strategy must specify an action for *every* node a player controls.

- Man's strategies: $\{F, D\}$
- Woman's strategies: Specify action at W_1 and action at W_2. E.g., FD means "F if he plays F, D if he plays D". Total strategies: $\{FF, FD, DF, DD\}$.

Normal Form Representation:

$M \setminus W$	FF	FD	DF	DD
F	(2, 1)	(2, 1)	(0, 0)	(0, 0)
D	(0, 0)	(1, 2)	(0, 0)	(1, 2)